Calyber: A Shared Rides Pricing and Matching Game

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1 Background and Introduction

Ride-sharing platform Calyber offers shared rides services to riders in Chicago. This is a carpooling service that matches two riders that have similar routes together to share a ride, thereby reducing the driving cost as well as offering lower prices to riders. Promoting shared rides is crucial for a sustainable future of urban transportation, reducing traffic congestion and carbon emission.

As we discussed in class, a key challenge in operating ridesharing platforms is to price and match incoming riders. For shared rides, these are more complicated as riders must be matched with each other in addition to being assigned a driver. While we primarily focused on the matching challenge in class, this game emphasizes pricing, assuming there are always enough drivers available where needed.

Riders arrive randomly over time. The platform sets a price for each arriving rider, and rider can choose to accept or reject the price. If accepted, the platform will try to match the rider with another rider, allowing them to share a ride to their destinations. The platform earns revenue from the total prices paid by them while covering the driving cost of picking them up and dropping them off. If no suitable match is available, the platform can defer the matching. However, riders have a limited waiting patience, and if exceeded, the platform misses the matching opportunity and must dispatch the rider solo, leading to higher costs.

In this game, you will play the role of the platform. The goal is to maximize profit by setting the price for each incoming rider as well as matching riders who accepted the prices (we call them requests or riders converted). The total profit is calculated as the total revenue collected from all riders minus the total driving costs of serving them. But here's the challenge—your decisions come with tricky trade-offs: for pricing, you have to properly assess the potential cost of serving the rider as well as balancing rider's willingness to pay. Assessing the cost is nontrivial because it depends on the matching outcome which might be uncertain at the time of rider arrival. For matching, the key trade-off involves deciding whether to match immediately or wait for future riders who may align better with the route. However, extended waiting times risk exceeding rider' patience thresholds, resulting in low-efficiency solo rides.

1.1 Definitions of Pricing and Matching Policies

We first define the pricing and matching policies. These policies are mappings from the system **state s** to pricing and matching decisions respectively. The state $\mathbf{s}$ is defined as **the set of waiting requests** (that is, the riders who have already accepted to pay the prices but are not matched and dispatched yet). The pricing and matching policies are thus defined as:

- **Pricing policy:** Given a state $\mathbf{s}$ and a rider i, the pricing policy $\psi_i(\mathbf{s})$ determines the price $p_i \in [0, 1]$ offered to rider i arriving at state $\mathbf{s}$.
- **Matching policy:** Given a state $\mathbf{s}$ and a rider i that converts at state $\mathbf{s}$, the matching policy $\phi_i(\mathbf{s})$ decides which waiting request $j \in \mathbf{s}$ to be matched with request i, or does not match and lets rider i wait first.

1.2 Game Dynamics

We proceed to describe the dynamics of the game.

Dynamics from the rider's perspective. The dynamics from the rider's perspective is described below and depicted in Figure 1. Imagine that a rider i arrives on the platform entering her origin O_i and destination D_i. The platform then quotes an upfront price $p_i \in [0, 1]$ (\$/mile). Based on the quoted price p_i and the rider's own willingness to pay $v_i \in [0, 1]$ (\$/mile) (the value of v_i is private and drawn from some distribution), this rider decides whether to accept the price and make a request (i.e., convert) or to leave the platform directly (i.e., not convert). Specifically, if $v_i \geq p_i$, then she converts; otherwise, she does not. Assume that ℓ_i is the trip length (in miles) if rider i rides solo. Then the amount she pays to the platform is $p_i \ell_i$.

For rider i just converted, the platform makes the matching decision. If the platform decides to match rider i with another waiting request, then the two riders are sent on a shared ride immediately. The platform can also choose to let rider i wait in the system first. She will stay in the system for a limited amount of time following an *exponential* distribution with a rate $\theta_i > 0$. If she is not matched with another rider before her patience explodes, the platform will have to dispatch her solo in an individual ride. Every time a ride is dispatched, the platform needs to pay a driving cost at a rate of c (\$/mile). Assuming ℓ_{ij} is the total length of a shared ride involving rider i and j (and let $\ell_{ii} = \ell_i$), then the driving cost is $c\ell_{ij}$ if it is a shared ride involving rider i and j, and the cost is $c\ell_i$ if it is an individual ride for rider i. Please see Section 4.3 for how ℓ_{ij} is calculated.

Dynamics from the platform's perspective. We also describe the dynamics from the platform's perspective and illustrate it in Figure 2. Assume that the system is in state $\mathbf{s}$ at some time t. Then there are two possibilities for the next event: either a new rider i arriving, or a waiting request $i' \in \mathbf{s}$ reneges. We discuss these two cases.

1. When the next event is rider i arriving, then the platform needs to determine the price p_i according to the pricing policy $\psi_i(\mathbf{s})$.
    - If rider i does not convert, then she leaves immediately. Nothing happens and state $\boldsymbol{s}$ does not change.

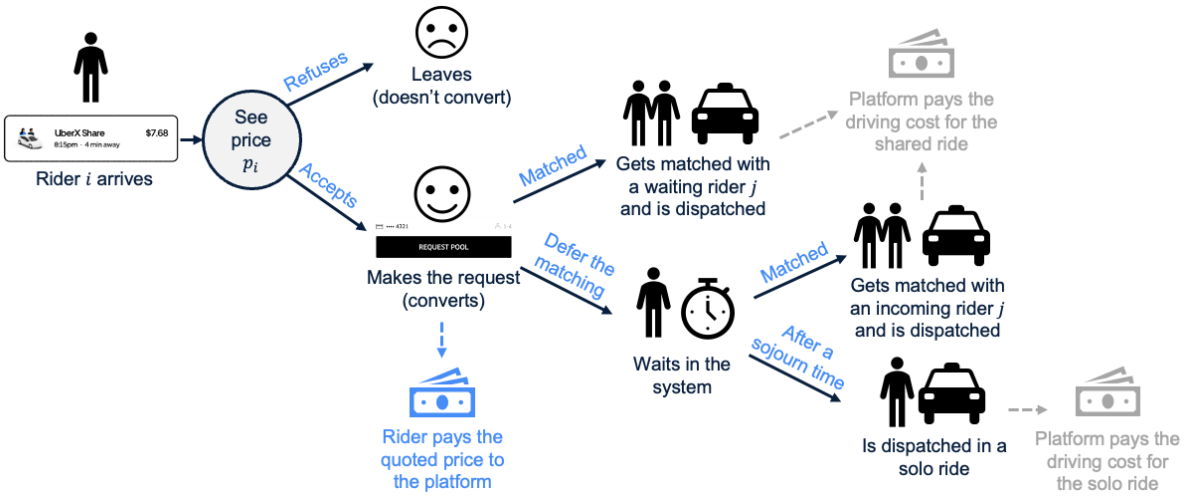


Figure 1: Dynamics from the rider's perspective.

- If rider i converts, then the platform collects a revenue of $p_i \ell_i$ and determines its match according to the matching policy $\phi_i(\mathbf{s})$. If the platform decides to match rider i with an waiting request $j \in \mathbf{s}$, then the platform pays a dispatching cost of $c\ell_{ij}$, and the state is updated to be $\mathbf{s} \backslash \{j\}$. If instead the platform decides not to immediately match rider i with any currently waiting requests, then she waits in the system and the state is updated to be $\mathbf{s} \cup \{i\}$.

2. When the next event is a request $i' \in \mathbf{s}$ running out of her patience, then she is dispatched solo and the platform pays a dispatching cost of $c\ell_i$. The state is updated to $\mathbf{s} \backslash \{i\}$.

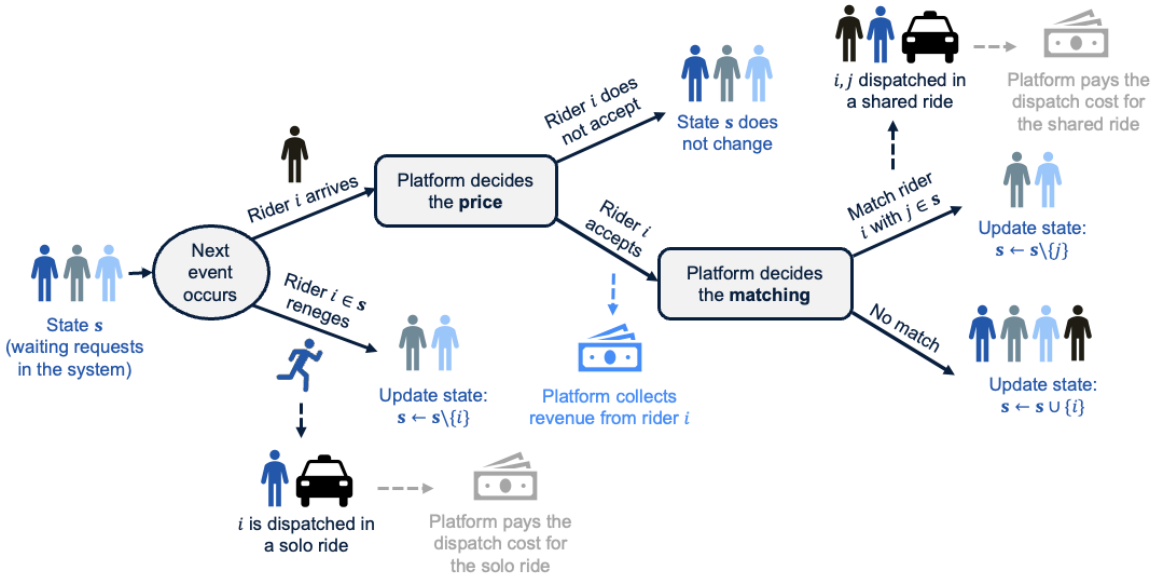


Figure 2: Dynamics from the platform's perspective.

2 Project Description

This section is a detailed overview of the project: what you are given, what you need to do, and how your policies will be evaluated. All relevant files are in the GitHub repository, which you can access by accepting the assignment at the Github Classroom link `https://classroom.github.com/a/-hKd-2L2`. Once you create or join a team, you'll see your team's repository, where you'll also submit your work.

2.1 What you are given

You are given historical rider arrival and request data, together with information on Chicago community areas. The driving cost rate is fixed at $c = 0.7$ ($/mile) throughout the game.

The student codebase includes the main notebook `student_policies.ipynb`, helper functions in `utils.py`, a demonstration notebook `helper_functions_demonstration.ipynb`, the rider class definition in `rider.py`, and the data files under the `data/` directory.

2.1.1 Training data

The training data contains information on 11,788 riders who arrived at the platform in the past. The data span a one-hour time window across six weeks. For each rider, the data include arrival time, pickup and dropoff information, and historical outcomes under a legacy platform policy.

The data are stored in `data/training_data.csv`. The first row is the header, and each subsequent row represents one arriving rider. The fields are:

- `rider_id`: Rider's unique ID (integer, 0–11787).
- `arrival_week`: Week of rider arrival (integer, 1–6).
- `arrival_time`: Arrival time within the hour (in seconds, 0–3600).
- `pickup_area`, `dropoff_area`: Pickup and dropoff community area indices (1–76).
- `pickup_lat`, `pickup_lon`: Exact latitude and longitude of the pickup location.
- `dropoff_lat`, `dropoff_lon`: Exact latitude and longitude of the dropoff location.
- `solo_length`: Solo ride distance (in miles), computed as the haversine distance between pickup and dropoff.
- `quoted_price`: Historical quoted price per mile ($/mile).
- `convert_or_not`: Whether the rider converted (1 for yes, 0 for no).
- `waiting_time`: Rider waiting time (in seconds), equal to dispatch time minus arrival time; `nan` if the rider did not convert.
- `matching_outcome`: Matched rider's ID if matched, otherwise `nan`.

2.1.2 Area information

Chicago is divided into 76 community areas. You are given the centroid latitude and longitude of each area in `data/community_areas.csv`. Each row represents one area. The columns are:

- `area`: Community area index (from 1 to 76).
- `lat`, `lon`: Latitude and longitude of the area centroid.

For optional clustering, visualization, or feature engineering, you may join the training data with the community area data through `pickup_area` or `dropoff_area`.

2.2 What you need to do

You need to implement your own pricing and matching policies in `student_policies.ipynb`. Please follow the instructions below.

1. Before writing code, your team should choose a unique team name. Suppose your team name is "Awesome Team".
2. Set the `TEAM_NAME` constant to your team name in camel case, for example `AwesomeTeam`.
3. Implement your policy logic in the provided notebook:
   - `pricing_function(state, rider)` should return a quoted price as a float.
   - `matching_function(state, rider)` should return either a rider object from the current state to match with, or `None` if the incoming rider should wait.
4. Run the provided tests in the notebook to verify that your implementation satisfies the required input/output format and does not have obvious bugs.
5. Export `student_policies.ipynb` into a python file named `TEAM_NAME_Policies.py` using the export function provided in the notebook / helper utilities.

Several additional important notes:

- **Time limit:** there are strict time limits on how long your policies are allowed to take for each decision. The execution time limit is **0.05 seconds per call** to the pricing or matching policy. If the time limit is exceeded, the algorithm is terminated, and the pricing decision defaults to $p = 1$, while the matching decision defaults to no immediate match. Therefore, any heavy training or model fitting should be done offline rather than inside these policy functions.
- **Packages:** you may not use additional Python packages beyond those already loaded in `student_policies.ipynb`. However, you may train models offline using other tools and then paste the resulting parameters or lightweight logic into the required functions.

- **Code format:** do not change the function signatures of `pricing_function` and `matching_function`. Your outputs must follow the format specified in the notebook template.
- **Helper functions:** helper functions are provided in `utils.py`. In particular, `populate_shared_ride_lengths()` computes shared-ride route statistics, and `test_policies()` can be used to validate your implementation. Examples are shown in `helper_functions_demonstration.ipynb`.
- **Testing examples:** pre-generated test examples are included to help you verify that your implementation behaves correctly before submission.

The repository also includes a benchmark policy folder, `Thea_policy/`, containing Thea's benchmark implementation, pre-trained models, and a training notebook for reference.

2.3 How your policies will be evaluated

Your policies will be evaluated by simulation on held-out test data. In addition to the training data, there will be an optional validation round and a final test round. The final test dataset contains three weeks of data collected from the same time-window structure as the training data. The simulation logic used for evaluation is shown in the pseudocode in Section 4.4, which closely follows the dynamics described in Section 1.2 and Figure 2.

Your policies will be ranked according to the total profit collected on the final test set. If several policies produce similar profit values, we may also consider the following additional metrics when determining the final ranking.

- `throughput`: total number of converted riders;
- `match_rate`: fraction of converted riders who are matched;
- `conversion_rate`: fraction of all arriving riders who convert;
- `cost_efficiency`: a normalized measure based on total dispatching cost;
- `avg_quoted_price`: average quoted price ($/mile) over all arriving riders;
- `avg_payment`: average payment ($/mile) over converted riders;
- `avg_waiting_time`: average waiting time (seconds) over converted riders.

3 Project Logistics and Timeline

3.1 Logistics

You can work on the project in a team of *at most* three people, the same team you worked with for the Sport Obermeyer case. Please pick a team name (make sure it's different from other teams). Your `TEAM_NAME_Policies.py` (exported from `student_policies.ipynb`) is the only file you need to submit for evaluation. You can only submit one python file

for each team. You will be ranked based on the performance of the test data set. Top teams will earn bonus scores and be invited to present how you develop your policies in the final lecture of the semester. Each team needs to submit a project report of 10 pages detailing your strategy.

3.2 Validation

To help you test and improve the performance of your policies, we provide an optional validation run before the final evaluation. Your code will be run on a separate validation dataset, and feedback will be returned to your team. This feedback may include a csv file with rider-level simulation outputs, including timeout information, as well as a csv file summarizing performance metrics. A validation ranking may also be produced across teams. The validation run is optional, though highly encouraged, and its performance will *not* affect your final score.

3.3 Timeline

- Week 8: Calyber game released.
- Week 12: (Optional) Submit your code for validation to your team's repository.
- Week 13: Submit your code for the final evaluation to your team's repository.
- Week 14: Ranking announced.
- Week 14: Top teams present their projects in the class.
- Final Week: Submit the project report.

3.4 Staff

If you have any questions regarding the codebase, contact Kenny Wongchamcharoen (pattaraphon.kenny@berkeley.edu), the GSI, or the professor for questions regarding developing your matching and pricing policies.

4 Appendix

4.1 Notations

Technical notation used in this document are listed here.

- O_i: the origin of rider i.
- D_i: the destination of rider i.
- v_i: the willingness-to-pay (\$/mile) of rider i. $v_i \in [0, 1]$.
- θ_i: the reneging rate of rider i.
- ℓ_i: the trip length (in miles) of the individual ride of rider i, i.e., the haversine distance between O_i and D_i.
- ℓ_{ij}: the shortest total length of the shared ride between rider i and j. $\ell_{ii} = \ell_i$.
- $\tilde{\ell}_{ij}$: the length of the shared part in the shared ride between rider i and j.
- c: the dispatching cost of rides (\$/mile).
- p_i: the price (\$/mile) quoted to rider i. $p_i \in [0, 1]$.
- $\mathbf{s}$: the state of the system, which is the set of waiting requests.
- $\psi_i(\mathbf{s})$: pricing policy.
- $\phi_i(\mathbf{s})$: matching policy.

4.2 Benchmark and Starter Materials

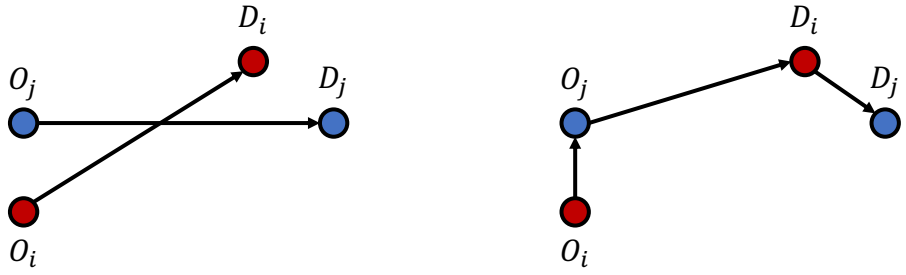
The repository includes starter materials to help you understand the game mechanics and implement your own policies. In particular, `student_policies.ipynb` provides the required policy interface, while `helper_functions_demonstration.ipynb` illustrates how to use key helper functions for routing and visualization.

In addition, the repository includes `Thea_policy/`, which contains Thea's benchmark policy. This benchmark is provided both as a learning reference and as a baseline for comparison during evaluation. The folder includes pre-trained machine learning models, the feature encoding objects used by those models, a training notebook (`Thea_policy_training.ipynb`) showing how the benchmark was developed, and the full implementation in `Thea_policy.py`.

You are encouraged to study Thea's policy to better understand how data-driven prediction and decision-making can be integrated into a pricing and matching strategy. However, your team is expected to design and submit its own original policy.

4.3 A Helper Function

The helper function `populate_shared_ride_lengths(origin_i, destination_i, origin_j, destination_j)` stored in `utils.py` calculates the driving cost and cost allocation of the shared ride involving any two riders i and j. For example, there are two riders i, j as illustrated in Figure 3a, where the origin-destination pair of rider i is (O_i, D_i) and that of rider j is (O_j, D_j). If they are matched together, the shortest distance is $O_i - O_j - D_i - D_j$ as in Figure 3b. The total length of this sequence is ℓ_{ij}.



(a) Not matched (b) Matched

Figure 3: Example of the shortest path and cost allocation of a shared ride.

The input of the function includes four arguments:

- `origin_i`: a tuple `(lat, lon)` indicating the latitude and longitude of O_i.
- `destination_i`: a tuple `(lat, lon)` indicating the latitude and longitude of D_i.
- `origin_j`: a tuple `(lat, lon)` indicating the latitude and longitude of O_j.
- `destination_j`: a tuple `(lat, lon)` indicating the latitude and longitude of D_j.

The output includes five terms:

- `trip_length`: the shortest length of the ride involving rider i and j (in miles). For example, in Figure 3b, it is the length of $O_i - O_j - D_i - D_j$. This is denoted by ℓ_{ij}.
- `shared_length`: the length that is shared by the two riders. For example, in Figure 3b, it is the length of $O_j - D_i$. This is denoted by $\tilde{\ell}_{ij}$.
- `i_solo_length`, `j_solo_length`: the length that is only taken by rider i (or j). For example, in Figure 3b, `i_solo_length` is the length of $O_i - O_j$, and `j_solo_length` is the length of $D_i - D_j$.
- `trip_order`: an integer from 0 to 4 denoting the the pickup and dropoff order of the shared ride in the shortest path. Define five constants `IIJJ=0`, `IJIJ=1`, `IJJI=2`, `JIJI=3`, `JIIJ=4`. For example, in Figure 3b, the optimal pickup-dropoff order is to pick up i - pick up j - drop off i - drop off j, so the `trip_order=IJIJ=1`.

4.4 Simulation Logic

The simulation logic of how your policies are evaluated on a data set is shown below.

Algorithm 1 Simulation

Require: Pricing policy $\psi_i(\mathbf{s})$, matching policy $\phi_i(\mathbf{s})$, driving cost rate c, set of arriving riders (where each rider i has features of origin O_i, destination D_i, willingness-to-pay v_i, and patience)

`profit` $\leftarrow 0$ ▷ Initial profit.
for each week in the data set **do**
  $t \leftarrow 0$ ▷ Accumulated time during a time window.
  $\mathbf{s} \leftarrow \emptyset$ ▷ Initial system state.
  repeat
    Receive the next event: a rider i arrives, or arequest $i \in \boldsymbol{s}$ reneges
    $t \leftarrow$ The time when the next event happens
    if next event is rider i arriving **then**
      Quote price $p_i = \psi_i(\mathbf{s})$ ▷ Use *pricing policy* to decide the quoted price.
      if $v_i < p_i$ **then** rider i does not convert
      else
        Rider i converts
        `profit` $\leftarrow$ `profit` $+ p_i \ell_i$ ▷ Collect the revenue from the request.
        $j \leftarrow \phi_i(\mathbf{s})$ ▷ Use *matching policy* to decide the request to match.
        if j is `None` **then** ▷ No match.
          $\mathbf{s} \leftarrow \mathbf{s} \cup \{i\}$
        else ▷ Match i with j.
          $\mathbf{s} \leftarrow \mathbf{s} \backslash \{j\}$
          `profit` $\leftarrow$ `profit` $- c\ell_{ij}$ ▷ Deduct the dispatching cost of the shared ride.
        end if
      end if
    else if next event is a request $i \in \mathbf{s}$ reneging **then**
      $\mathbf{s} \leftarrow \mathbf{s} \backslash \{i\}$
      `profit` $\leftarrow$ `profit` $- c\ell_i$ ▷ Deduct the dispatching cost of the individual ride.
    end if
  until $t > 3600$ ▷ The one-hour time window closes.
  if $\mathbf{s} \neq \emptyset$ **then**
    `profit` $\leftarrow$ `profit` $- c\ell_i$ ▷ Remaining requests are all dispatched solo.
  end if
end for